Status	Finished
Started	Thursday, 8 October 2026, 9:36 AM
Completed	Thursday, 8 October 2026, 9:45 AM
Duration	8 mins 39 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted

b. The search stops

c. The search continues in the left or right sublist ✓

d. The search continues from the beginning

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. Number values should only be present

  Number values should only be present
- c. There should be direct access to the middle element in any sublist
- d. The list must be sorted

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search works on unsorted lists
- b. Binary search has a lower time complexity on large, sorted lists ✓
- c. Binary search does not require dividing the list
- d. Binary search can find multiple instances of the target element

Question **4**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Divide the list into two equal parts
- c. Compare the target element with the middle element in the list ✓
- d. Sort the list

Question **5**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Linear
- c. Hash
- d. Both (1) & (3)

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By using a hash function
- b. By comparing each element in the list sequentially ✓
- c. By sorting the list first
- d. By dividing the list into halves

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. The last element is returned
- c. An error is raised
- d. The search is considered unsuccessful ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Hash search

c. Linear search ✓

d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O👎

c. O(n log n)

d. O(log n) ✓

Question **10**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Complexity

c. Insertion

d. Bubble ✓

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops

b. The lower middle element is chosen as the middle element ✗

c. The middle element is chosen randomly

d. The higher middle element is chosen as the middle element

Question **12**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is not the array at all

c. When the item is the last element in the array

d. When the item is somewhere in the middle of the array ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94

b. 90 and 99 ✓

c. 90 and 94

d. 89 and 99

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Linear search ✓

b. Interpolation search

c. Hash search

d. Binary search

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n

b. n/2

c. n ✓

d. n-1